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Stochastic Strategic Demand-Capacity Balancing Using Model-Free Reinforcement Learning

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The strategic role of ATFM

Strategic demand–capacity balancing is essential to mitigating inefficiencies in air traffic operations.

However, this is a challenging process, particularly due to the dynamic and uncertain nature of weather conditions.

74.3%

of U.S. system-impacting delays over 15 min
are weather-related

FAA, 2017–2023

20–50%

of European ATFM delay is attributed to
weather

EUROCONTROL, 2024

40.6%

of ATFM measures applied in the Brazilian
airspace due to weather

DECEA, 2025



The strategic role of ATFM

Increasing complexity of air traffic operations.

The increasing density and diversity of air traffic operations introduce new challenges for demand-capacity balancing.

Urban Air Mobility (UAM) presents unique operational characteristics

- 01 Very short flights, so delay decisions must resolve within minutes
- 02 Distributed vertiports with limited parking positions
- 03 Tight aircraft turnaround coupling arrivals to departures
- 04 Short-notice service expectations from passengers
- 05 Non-invertible capacity envelopes (a minimum departure count per arrival level)



Optimization approaches for ATFM decision support

From model-based deterministic approaches to learning-based approaches under uncertainty.

DETERMINISTIC GHP

Richetta & Odoni (1994), Bertsimas et al. (2011): foundational, but forecast-dependent and without tactical integration.

STOCHASTIC ATFM

Estes & Ball (2020), Chen et al. (2017), Shone et al. (2021): uncertainty modelled, strategic-tactical coupling still separate.

RL FOR ATFM

Jones et al. (2023), Taylor et al. (2023), Carvalho & Murça (2025): promising, but no unified strategic-tactical DCB across ATM and UAM.

The ground-holding problem—deciding which aircraft to hold on the ground to avoid more costly airborne delay—has driven decades of optimization research.

However, traditional approaches often rely on **deterministic forecasts and predefined decision schemes**, limiting their adaptability to stochastic demand-capacity variations and weather uncertainty.



Research Contributions

01 Integrated RL framework

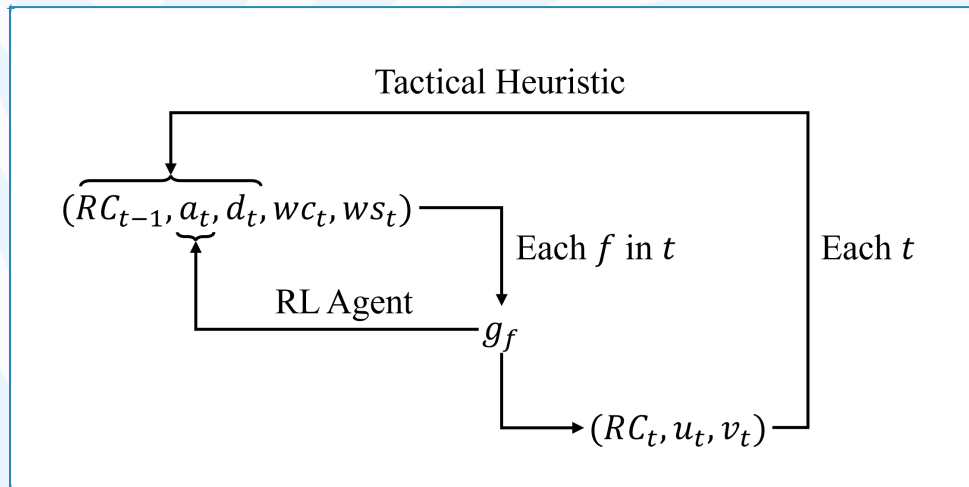
Develop a reinforcement learning framework for demand-capacity balancing that integrates strategic ground-delay allocation with tactical capacity management under uncertain weather forecasts.

02 Application to different operational paradigms

Demonstrate the applicability of the proposed RL framework to both conventional ATFM and UAM traffic management.



One loop, two decision layers



Strategic layer – RL agent

For each arrival flight f in epoch t , choose a ground delay g_f in 15-minute increments, up to one hour.

Tactical layer – heuristic

Given the wind state, select a feasible runway configuration and set arrival/departure service rates from the capacity envelope.

Coupling

Reallocated demand changes future queues, so the tactical consequence of each delay is fed back into the reward.

Decision epoch: 15 min | Horizon: $T = 96$ epochs (one day)

State: an MDP on 15-minute epochs

$$s_t = (RC_{t-1}, a_t, d_t, wc_t, ws_t)$$

RC_{t-1}

PREVIOUS CONFIGURATION

A, B, or unavailable

a_t

ARRIVAL QUEUE

Cumulative unserved arrivals

d_t

DEPARTURE QUEUE

Cumulative unserved departures

wc_t

WEATHER

VMC, IMC, LIMC

ws_t

WIND STATE

0-3: which sides are usable

$ws = 0$

Both configurations available (CW $\leq$ 20 kt, TW $\leq$ 5 kt)

$ws = 1$

Only 10R/10L

$ws = 2$

Only 28L/28R

$ws = 3$

None available



Action: per-flight ground delay, masked

ACTION SPACE

$$g_f \in \{0, 1, 2, 3, 4\} \times 15 \text{ min}$$

Capped at one hour, and at the time remaining in the horizon, to keep schedule-level reallocation operationally moderate.

DEMAND REALLOCATION

$$a_{t+g_f}^{\text{sched}} \leftarrow a_{t+g_f}^{\text{sched}} + 1$$

$$a_t^{\text{sched}} \leftarrow a_t^{\text{sched}} - 1$$

TACTICAL SERVICE RATES

$$u_t = \min(u_{\max} [wc_t, RC_t], a_t)$$

$$v_t = \min(v_{\max} [wc_t, RC_t, u_t], d_t)$$

Configuration chosen from the operationally valid set $RC_t \in RC(ws_t)$.

Two sources of stochasticity

1. WEATHER DYNAMICS – METAR

$$P(\text{METAR}_t = j \mid \text{METAR}_{t-1} = i)$$

Empirical transition frequencies between VMC, IMC and LIMC at 15-minute resolution.

2. FORECAST ERROR – TAF VS METAR

$$P(\text{METAR} = j \mid \text{TAF} = i)$$

Confusion matrices estimated by comparing forecasts against observed realizations.

TRAINING-TIME SAMPLING

$$wc_t \sim P(wc_t \mid wc_{t-1}, \hat{wc}_t) \propto P(wc_t \mid wc_{t-1}) \cdot P(wc_t \mid \hat{wc}_t)$$

The wind state ws_t is sampled the same way. Wind speed and direction are converted to crosswind and tailwind components relative to runway orientation, then discretized into the four wind states.



Cost of the tactical consequence, not just the strategic delay

TACTICAL COST

$$J^{\text{tac}} = \sum_{t=0}^T [\alpha(a_t - u_t)^2 + (d_t - v_t)^2 + \delta \cdot 1(RC_t \neq RC_{t-1}) \cdot (\alpha a_t^2 + d_t^2)]$$

α weights airborne against ground delay; δ prices configuration switching.

STRATEGIC COST

$$J^{\text{strat}} = \eta \sum_f g_f$$

$\eta = 0.1$

REALLOCATION EFFECT

$$\Delta J^{\text{tac}} = J^{\text{tac}}_{\text{delayed}} - J^{\text{tac}}_{\text{original}}$$

Over the remaining episode

STEP REWARD

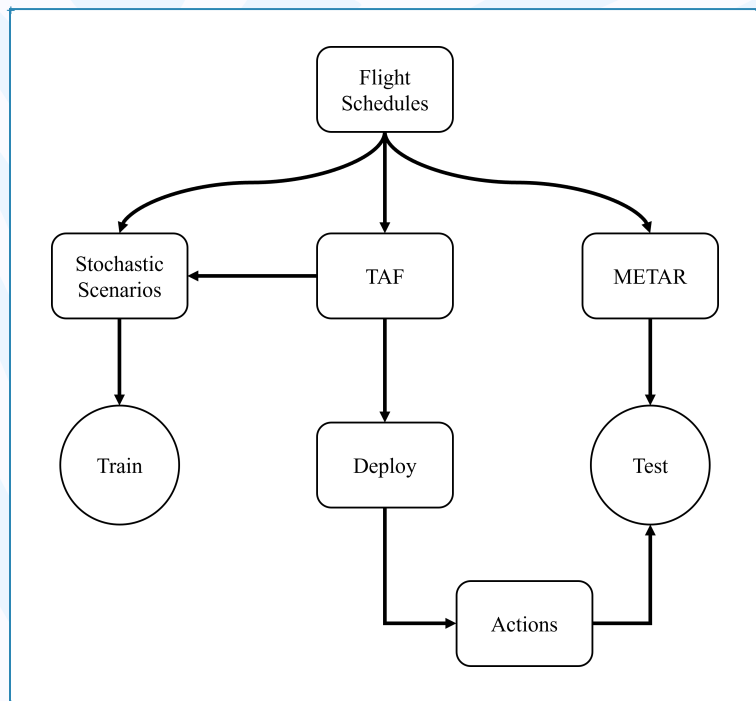
$$R_t = -(\Delta J^{\text{tac}}(t) + \eta \sum_{f=1}^{F_t} g_f)$$

A delay may relieve congestion in one future epoch while worsening another. Because the reward carries the full change in tactical cost, the policy has to balance the two rather than simply minimizing delay minutes.

Evaluation metric: €45 per delayed departure minute, €90 per airborne arrival minute, following EUROCONTROL (2024).



Train on forecast, test on observation



ALGORITHM

Maskable PPO

Stable Baselines 3 · Gymnasium

EPISODES / DAY

500,000 conventional

300,000 UAM

HYPERPARAMETERS

$lr\ 3 \times 10^{-4}$ · batch 64 · entropy 0.01 · clip 0.2

TEST SET

30 randomly selected days
diverse weather and demand

BASELINE

No rescheduling: the same tactical heuristic and capacity model with $g_f = 0$ for every flight. In evaluation the strategic rescheduling cost is omitted, so the reported metric isolates tactical congestion reduction.

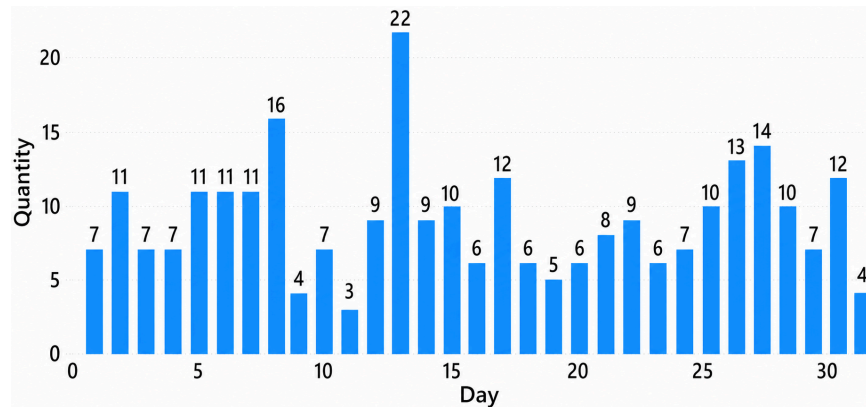
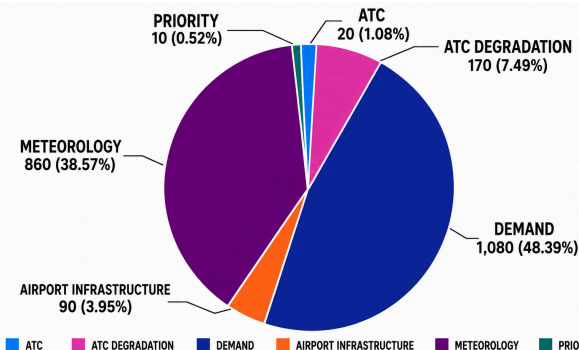
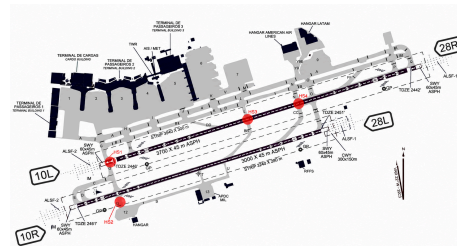
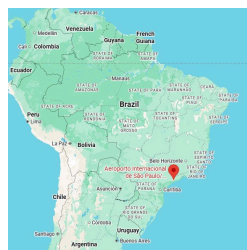
Hyperparameters were not exhaustively tuned; sensitivity to them remains a limitation.

Traffic Flow Management at SBGR

300k+

annual operations at São Paulo/Guarulhos (SBGR)

GRU Airport, 2025



Delay factors at SBGR, 2025 – meteorology 38.57%.

ATFM measures by day of the month, SBGR, January 2026.



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Data description

OBSERVED WEATHER

24 months

METAR, 2022–2023, at 15-min resolution: wind direction and speed, visibility, ceiling, IMC/LIMC flags. Source: REDEMET.

FORECASTS

TAF

Same variables at 15-min resolution, used to drive the stochastic training scenarios. Source: REDEMET.

FLIGHT DATA

40 NM

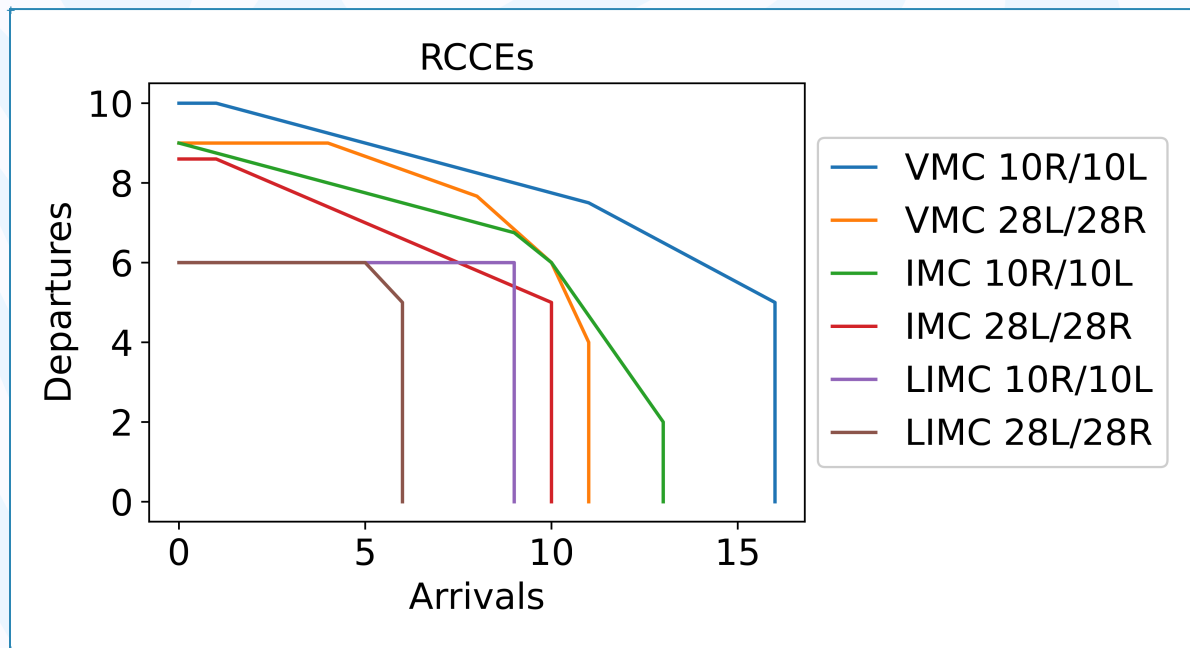
Arrival estimates plus scheduled and observed times. ANAC SIROS schedules corrected with ANAC VRA and FlightRadar24 tracks to remove buffers.

SPLIT

Transition and forecast-error probabilities are derived from the observation and forecast records. **TAF for training, METAR for testing**, over 30 randomly selected days with diverse weather and demand.



Runway capacity modeling



Envelopes fitted for 10R/10L and 28L/28R under each weather condition. Switching configuration carries an explicit cost in the reward.

Estimated from historical arrival-departure data by **quantile regression** (Ramanujam & Balakrishnan, 2009): a piecewise-linear frontier fitted by minimizing weighted deviations, with monotonically decreasing slopes and continuity at the breakpoints.

$$\min \sum_{n=1}^N Z_n$$

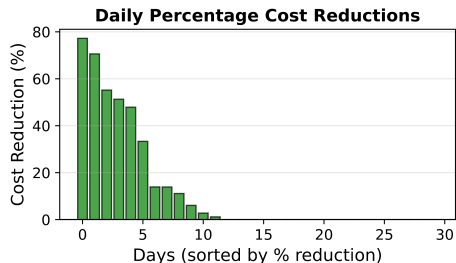
with deviations weighted by $\omega_p = (100 - p)/p$ and slopes $\beta_{k+1} \leq \beta_k$.

QUEUE DYNAMICS

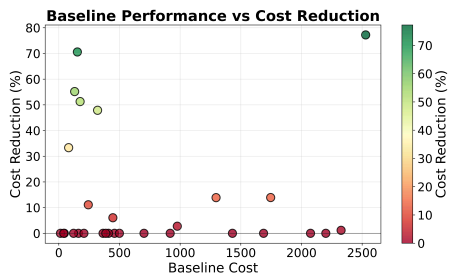
$$a_{t+1} = a_t + a_t^{\text{sched}} - u_t$$

$$d_{t+1} = d_t + d_t^{\text{sched}} - v_t$$

Performance evaluation



Daily percentage cost reduction against the baseline, 30 test days at SBGR.



Daily percentage reduction against baseline cost: the policy matches the baseline on most days, with large gains on a minority – including the most congested day (~77%).

13.3%

aggregate mean-cost reduction

STATISTICAL SIGNIFICANCE

Wilcoxon signed-rank $p = 0.002183$

Non-parametric test, appropriate for the non-normal cost distributions.

The SDCB distribution is narrower and concentrated at lower costs; the baseline shows longer tails from occasional severe congestion episodes.



Gains concentrate on congested days

12/30

days improved (40%)

77.2%

maximum daily reduction

0

days with degraded performance

1.1%

of flights delayed at all

Strategic ground delay allocation – 10,144 flights

ALL FLIGHTS

98.9% flown as scheduled · 1.1% delayed (10,144 flights, true scale)

DELAY	FLIGHTS	SHARE
0 slots	10,033	98.9%
1 slot (15 min)	68	0.7%
2 slots (30 min)	39	0.4%
3 slots (45 min)	0	0.0%
4 slots (60 min)	4	0.0%

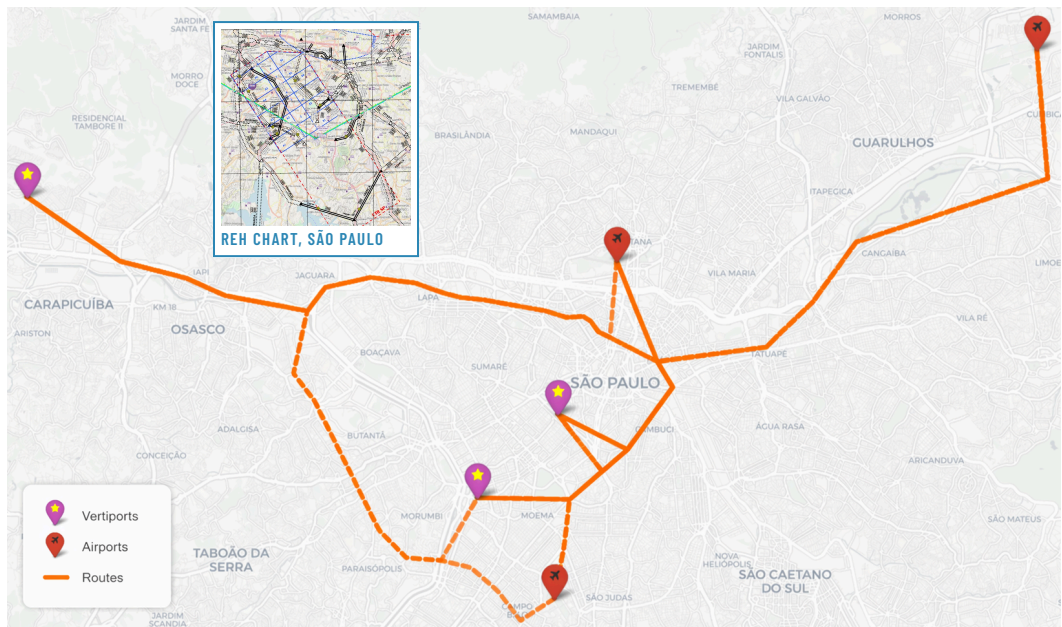
READING THE RESULT

The agent learns a **conservative** policy: almost all flights fly as scheduled, and delay is spent only where it removes real tactical congestion. That protects airlines from rescheduling cost while still cutting the upper tail of the cost distribution.

The 18 unchanged days had minimal baseline congestion — the correct action there is to do nothing, and the policy does exactly that.



UAM operations in the São Paulo Metropolitan Region (RMSP)



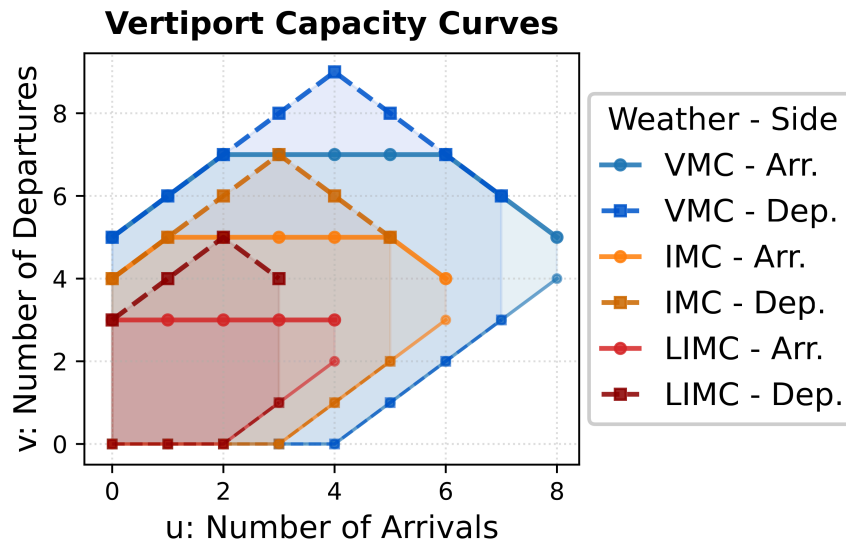
Three vertiport candidates from the helipad distribution across the RMSP, with optimal routes to SBGR, SBSP and SBMT (Carvalho, 2026).

DEMAND PIPELINE

1. Extract daily and 15-min slot demand from helicopter traffic data
2. Resample arrivals and departures independently
3. Generate flight records with duration variability from BlueSky eVTOL simulations on the REH route network
4. Pair with SBGR METAR and TAF

A deliberately demanding high-density scenario, not a forecast of mature air-taxi demand. The baseline is not pre-adjusted against vertiport capacity.

Vertiport capacity modeling



Side A is arrival-favouring (9 arrivals / 7 departures in VMC); Side B is departure-favouring (7 / 9). IMC and LIMC progressively reduce $U_{\max}$, $V_{\max}$ and $C_{\max}$.

Conventional airports allow zero departures at any arrival level. Vertiports do not: aircraft availability and limited parking impose a **minimum departure count** per arrival level, and handling capacity imposes a maximum.

$$\begin{aligned}
 0 &\leq u \leq U_{\max} [wc, \sigma] \\
 0 &\leq v \leq V_{\max} [wc, \sigma] \\
 u + v &\leq C_{\max} [wc, \sigma] \\
 u + d_{\min} [wc, \sigma] &\leq v \leq u + d_{\max} [wc, \sigma]
 \end{aligned}$$

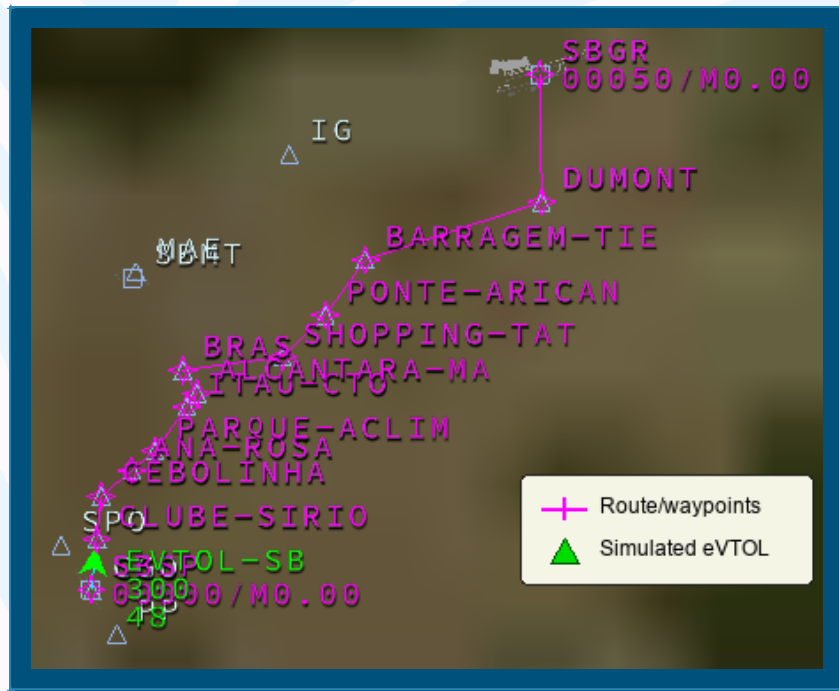
TACTICAL HEURISTIC, ADAPTED

Keep the previous side if still feasible; otherwise take the available side with the largest arrival capacity, then search downward until the departure queue covers the required minimum:

$$\begin{aligned}
 u_i &= \max\{u : u \leq \min(a_i, U_{\max} [wc_i, \sigma_i]), v_{\min} [wc_i, \sigma_i, u] \leq d_i\} \\
 v_i &= \min(d_i, v_{\max} [wc_i, \sigma_i, u_i])
 \end{aligned}$$

σ_t : operational side selected at epoch t .

UAM traffic simulation



A simulated eVTOL flight between **SBSP** and **SBGR** in the BlueSky simulator, following the published waypoints of the Helicopter Special Routes (REH).

Routes Existing REH corridors over the metropolitan region, reused for the eVTOL network

Output Realistic flight-duration variability for the synthetic schedule generator

EVTOL PROFILE

Max cruise speed **96 kt**

Cruise altitude **1,500 ft AGL**

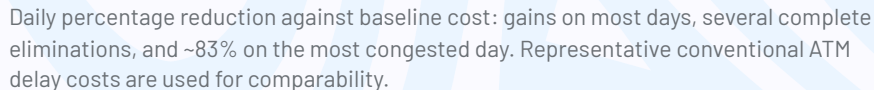
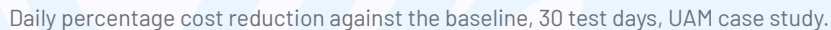
Intermediate step **300 ft AGL at 48 kt**

Parameters from Sarkar et al. (2025)

Route-level low-altitude weather is unavailable, so SBGR METAR/TAF stand in for conditions in the vertiport area.



Performance evaluation



aggregate mean-cost reduction

Wilcoxon $p = 0.000281$

Non-parametric test, appropriate for the non-normal cost distributions.

Several test days see the tactical cost eliminated entirely, and the baseline's long upper tail is absent under SDCB: maximum daily cost falls from €8730 to €1440.

More intervention, more benefit

23/30

days improved (76.7%)

43.9%

mean daily percentage reduction

4.4%

of UAM flights delayed – four times the conventional rate

Strategic ground delay allocation – 2,857 flights

ALL FLIGHTS

95.6% flown as scheduled · 4.4%
delayed (2,857 flights, true scale)

DELAY	FLIGHTS	SHARE
0 slots	2,731	95.6%
1 slot (15 min)	62	2.2%
2 slots (30 min)	50	1.8%
3 slots (45 min)	5	0.2%
4 slots (60 min)	9	0.3%

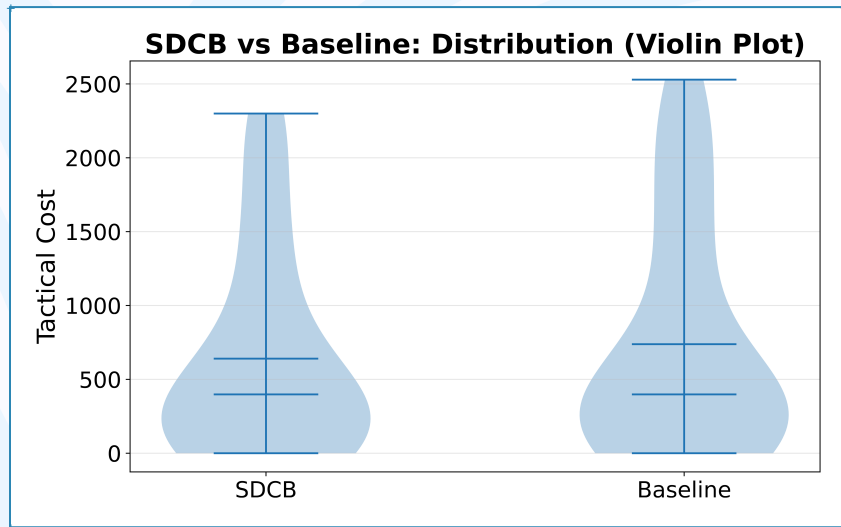
WHY THE DIFFERENCE

Conventional schedules are already operationally planned, so there is less slack to recover. The synthetic UAM baseline is deliberately not pre-optimized against vertiport capacity, leaving more headroom for strategic reallocation – and the coupled arrival/departure envelope makes that reallocation more consequential.



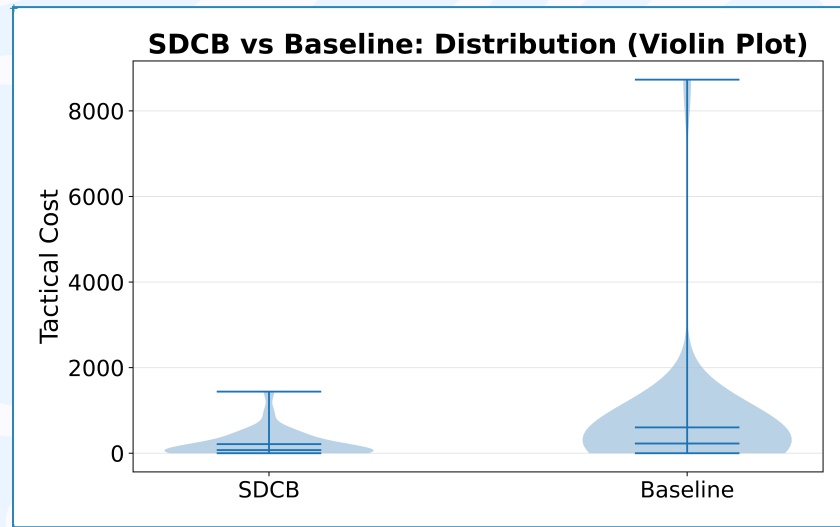
SDCB against the baseline, day by day

Case 1 – SBGR conventional ATM



Same median, narrower spread: IQR falls from 1061 to 847 and the upper tail is trimmed.

Case 2 – UAM in the São Paulo Metropolitan Region (RMSP)



The baseline tail reaching €8,730 is cut to €1,440, and the standard deviation falls from 1539 to 319.

Contributions and future work

Contributions

- 01 A model-free RL framework for strategic DCB that handles weather uncertainty through probabilistic weather representation and empirically grounded demand scenarios
- 02 Integrated strategic and tactical optimization: ground delay allocation coupled with configuration and service-rate decisions
- 03 Purpose-built MDP formulation linking traffic, weather and operational state to per-flight ground-delay actions, with tactical congestion consequences embedded in the reward.
- 04 Demonstrated transfer across conventional ATM and a UAM case study with the same algorithm

Limitations and next steps

SINGLE AIRPORT

Network effects, airline recovery, passenger connections, multi-airport coordination and wind-state definitions elsewhere remain unexplored.

WEATHER REPRESENTATION

VMC/IMC/LIMC categories from airport METAR/TAF; route-level low-altitude weather and finer variables would improve realism.



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Carvalho &
Murça (2025)



Carvalho et al.
(2026)